

Homework 5

Release Date: 1404/9/4

Deadline: 1404/9/12

(*) starred problems are optional and have a bonus mark!

1 FM Modulation (12 points)

The signal $m(t) = 20 \text{ sinc}(200t)$ is modulated in FM form using the carrier $c(t) = 100 \cos(2\pi f_c t)$ and the modulation index is equal to 6.

1. Find the modulated signal.
2. What is the maximum frequency deviation?
3. What are the power and the bandwidth (Carson's rule) of the modulated signal?

2 FM Modulator (12 points)

The signal $m(t)$ is applied to an FM modulator. The maximum frequency deviation is 25 Hz/V. Plot the phase and instantaneous-frequency curves.

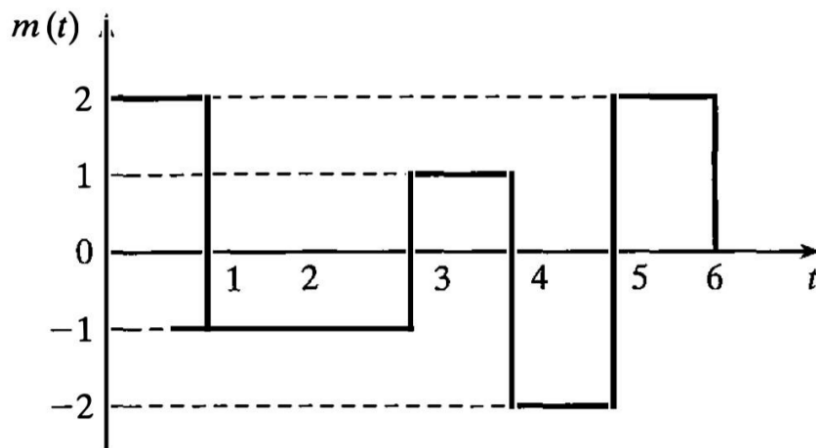


Figure 1: The signal $m(t)$.

3 Frequency Components of FM (12 points)

The signal $m(t) = 5 \cos(20000\pi t)$ is used to modulate the carrier $c(t) = 100 \cos(2\pi f_c t)$ into FM form. The peak frequency deviation is 20 kHz. The carrier frequency is $f_c = 100$ MHz.

1. Determine the magnitudes and frequencies of all FM components whose amplitudes correspond to at least 10% of the total signal power.
2. Using Carson's rule, obtain the approximate bandwidth.

4 FM Signal Power (25 points)

Follow the steps below to show the power of FM signal $x(t) = A_c \cos(2\pi f_c t + \phi(t))$ is $\frac{A_c^2}{2}$.

1. Write the power expression for the FM signal and show that the power equals

$$P = \frac{A_c^2}{2} + I,$$

where

$$I = \lim_{T \rightarrow \infty} \frac{A_c^2}{2T} \int_{-T/2}^{T/2} \cos(4\pi f_c t + 2\phi(t)) dt.$$

2. Show that

$$I_\infty = \int_{-\infty}^{\infty} \cos(4\pi f_c t + 2\phi(t)) dt$$

relates to the Fourier transforms $\mathcal{F}\{e^{j2\phi(t)}\}$ and $\mathcal{F}\{e^{-j2\phi(t)}\}$ at the frequencies $-2f_c$ and $2f_c$, respectively.

3. Use Taylor series expansion to show that I_∞ depends on the Fourier transforms $\mathcal{F}\{\phi^{(n)}(t)\}$ at the frequency $\pm 2f_c$.
4. Show that if $f_c \gg W$, where W is the bandwidth of the message-related phase $\phi(t)$, then $I_\infty \approx 0$.
5. Show that the power is approximately equal to $\frac{A_c^2}{2}$.

5 Envelope Detector for an FM Signal (15 points)

The FM signal

$$s(t) = A_c \cos\left(2\pi f_c t + 2\pi k_f \int_0^t m(\tau) d\tau\right)$$

is applied to the system shown below. The system consists of a high-pass filter (formed by the series capacitor C and resistor R), and an envelope detector. Assume the resistance R is small compared with the reactance of C over the bandwidth of $s(t)$, so that the envelope detector does not introduce distortion before detection. Find the output signal under the condition

$$k_f |m(t)| < f_c.$$

6 Frequency Spectrum of an FM Signal (12 points)

The message signal

$$m(t) = A \cos(2\pi f_1 t) + B \cos(2\pi f_2 t)$$

passes through an FM modulator and the FM signal

$$x_c(t) = A_c \cos(2\pi f_c t + \beta_1 \sin(2\pi f_1 t) + \beta_2 \sin(2\pi f_2 t))$$

is produced. Find the spectrum of $x_c(t)$ (The modulation indices are $\beta_1 = \frac{\Delta f_1}{f_1}$, $\beta_2 = \frac{\Delta f_2}{f_2}$).

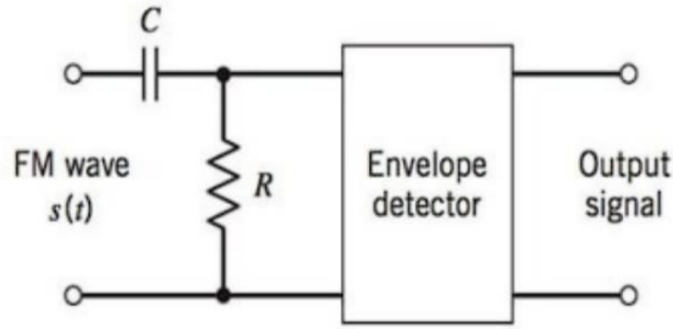


Figure 2: Envelope detector for an FM signal.

7 Stereo FM Broadcasting (12 points)

Many FM radio stations transmit music in stereo using two microphones placed at different locations. Investigate the stereo FM transmitter and receiver, draw the block diagram of each, and explain their operation.

8 *Superheterodyne Receiver (10 points)

A superheterodyne receiver operates over the frequency range

$$88 \text{ MHz} - 108 \text{ MHz}.$$

The intermediate-frequency (IF) and local-oscillator (LO) frequencies are chosen such that

$$f_{IF} < f_{LO}.$$

We require that the image frequency f'_c lie outside the band

$$88 \text{ MHz} - 108 \text{ MHz}.$$

Determine the minimum value of f_{IF} and the corresponding range of the local-oscillator frequency f_{LO} .

Submission

Please submit your answers in a .pdf file named `CS_HW5_studentID.pdf`, replacing `studentID` with your actual student ID.